



Universal Power Monitor - Head Setup and Calibration

([Click here for Summary Version](#))

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	30 mins	Not Applicable

2 Overview

Last Update	05/17/2007
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3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Power Measurement Kit (Wattmeter Kit)	1	-	2372868	-

3.2 Safety



**DANGER**

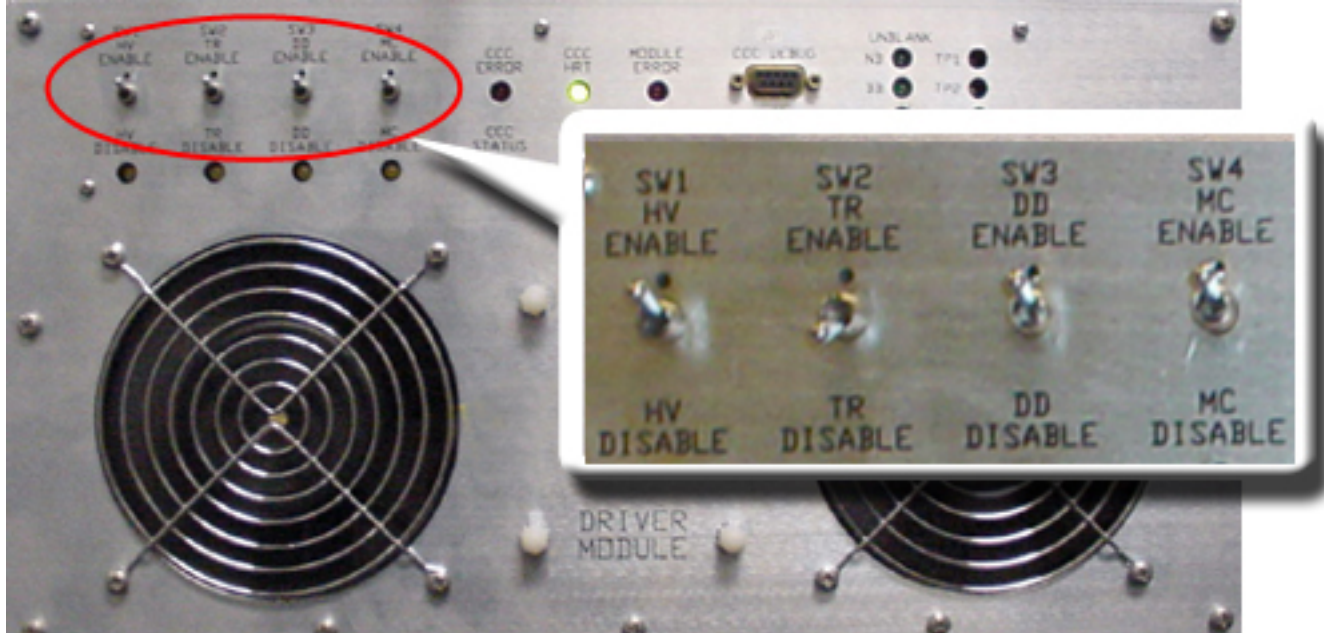
FATAL ELECTRIC SHOCK HAZARD!!
DANGEROUS VOLTAGES THROUGHOUT THE CABINET
TO PREVENT FATAL ELECTRIC SHOCK, MAKE SURE THE RF AMPLIFIER IS DISABLED WHEN CONFIGURING THE CALIBRATION TOOLS.

4 Procedure

4.1 HEAD FORWARD POWER CALIBRATION

1. Inhibit TR faults. Disable the Driver Module TR fault detection in the RFS (RF in the System Cabinet), (see [Illustration 1](#)). Move switch 2 to the TR Disable position.

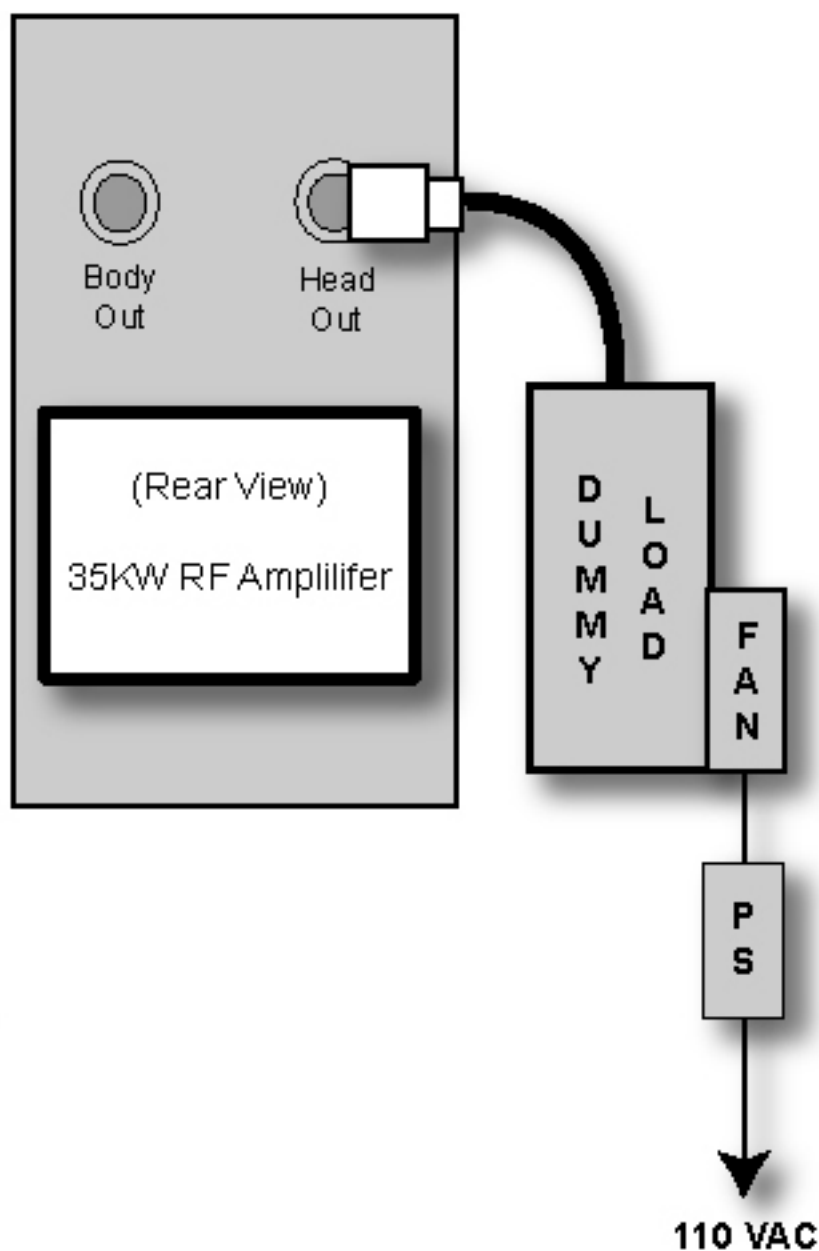
Illustration 1: Driver Module Front Switches



2. Remove the Head Heliax cable from J3. Connect directly to the 35KW RF Dummy Load. Plug the power supply into the Dummy Load and the power supply into 110 VAC service outlet in the System Cabinet. (see [Illustration 2](#)).

NOTE: If the RF Max power out procedure was just completed, configuration is the same. Leave the Body RF heliax cable disconnected and the head output can keep the coupler in line. It will not change the outcome of the calibration.

Illustration 2: Setup for Head Output Measurement

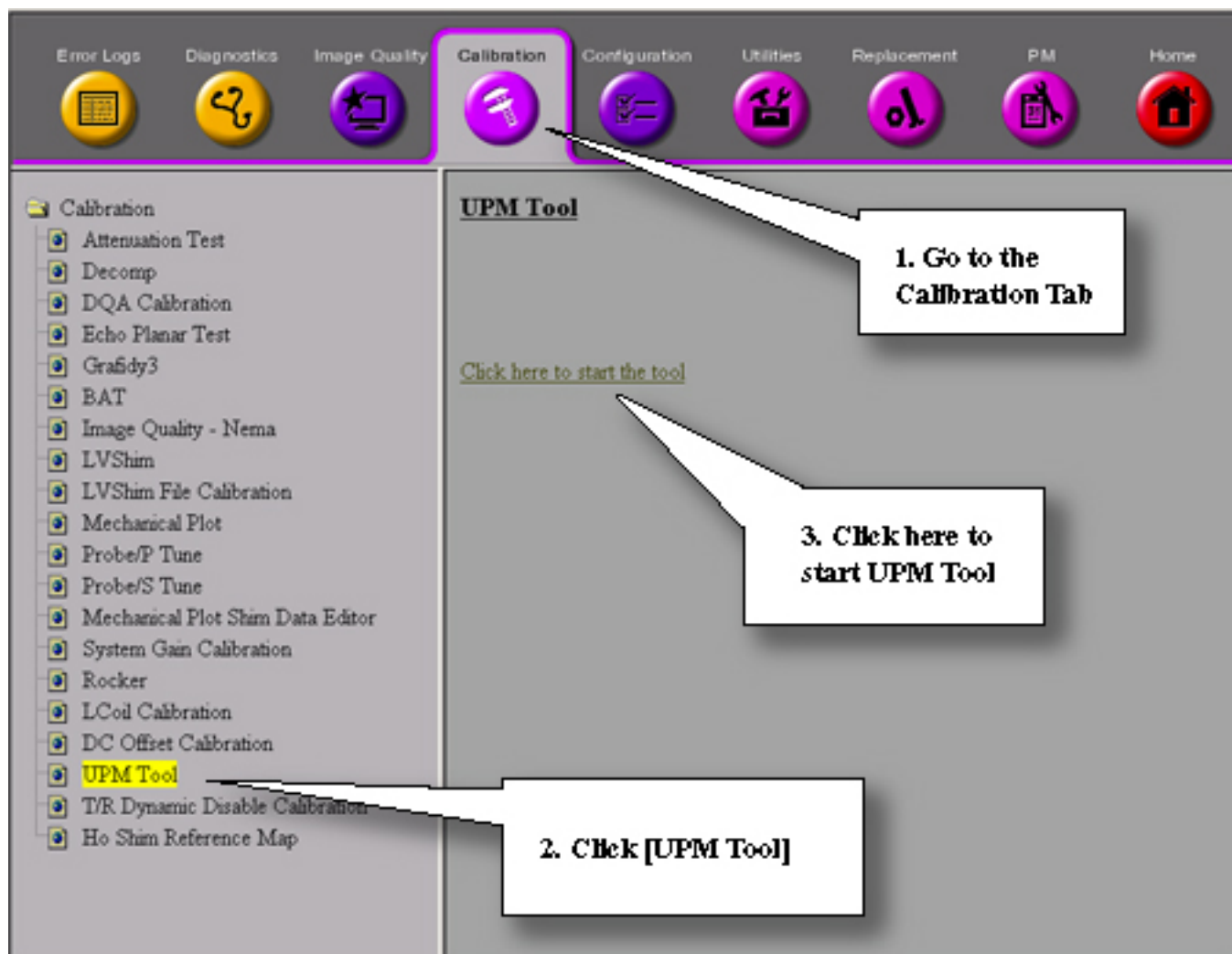


4.2 Setting up Software Tool to Calibrate UPM – Head Forward Power

NOTE: This tool must see a Landmark only. It is not necessary to have a Head coil in place, or any other scan parameter entered than what is described below. Simply Landmark on the cradle where the Head coil would be.

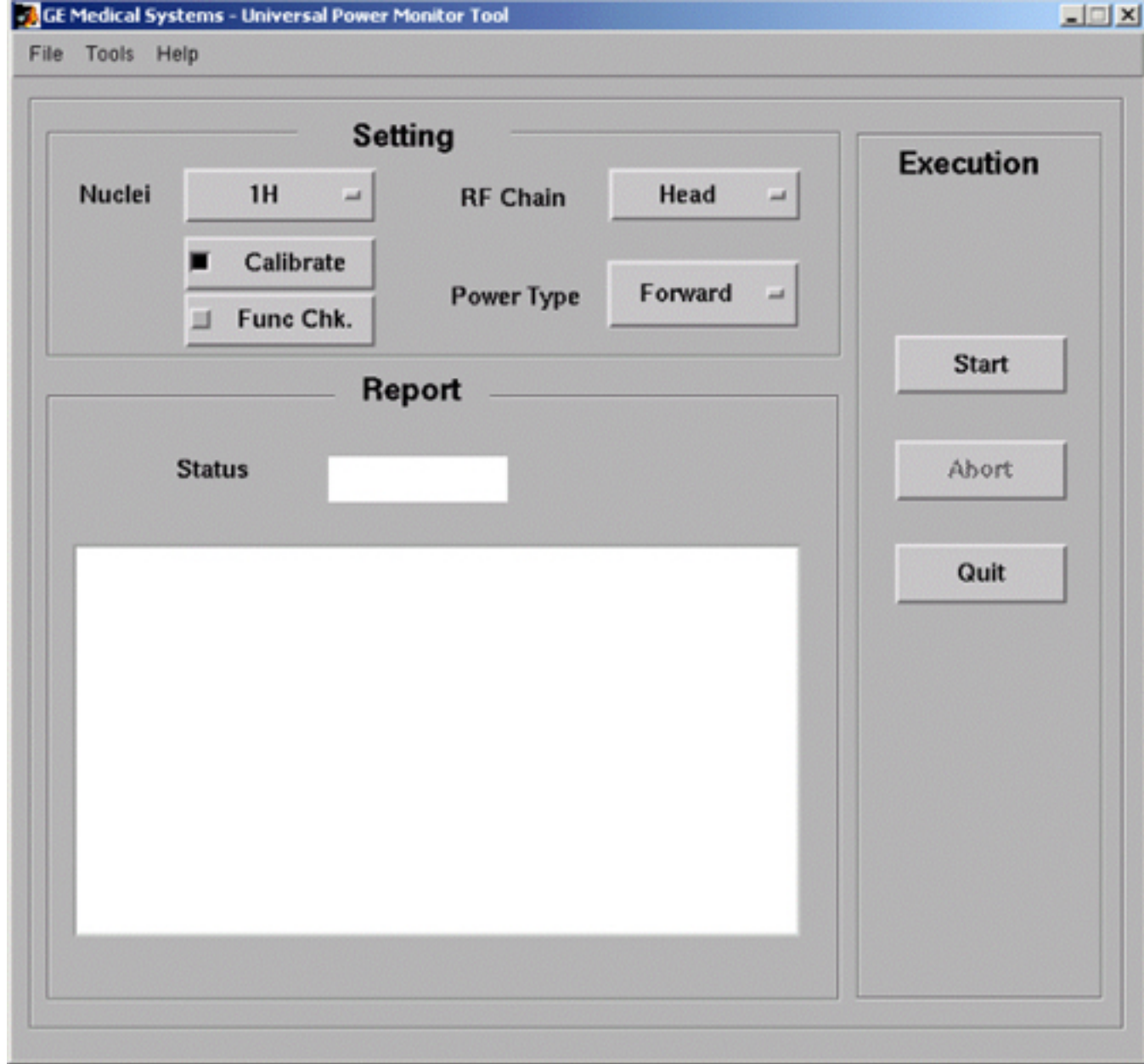
1. To establish a Landmark, Start a [New Exam].
2. Enter the following parameters:
 - a. Patient ID: **geservice**
 - b. Weight: **111**
3. Press [Align] button on.
4. Move the cradle to the Home position and then drive it into the bore.
5. Press **Landmark** button when the align light is approximately where the center of the head coil would be. Press the **Advance to Scan** button.
6. From the Common Service Desktop Select:
 - a. Select [Calibration Tab]
 - b. Select [UPM Tool]. [Illustration 3](#).

Illustration 3: Starting UPM Tool



- c. Select [Click here to start the tool] to start UPM tool. See [Illustration 3](#).
7. Select the Nuclei [1H]. See [Illustration 4](#).

Illustration 4: UPM Monitor Tool GUI



NOTE: The first time running the UPM Calibration tool after a load from cold, the following Error may appear: “The spectroRFampType and upm.config do not match! Please run the UPM Tool config file editor and hit [Restore Defaults] and save the config file”. If this error appears, click [File] -> “Edit Config”. When the window opens, click [Restore Defaults] then close the screen. The tool will now have the correct information and this pop-up will not appear again during future UPM Calibrations.

8. Select the RF Chain you are going to calibrate [Head].
9. Select: [Calibrate] Button
10. Select Power Monitor type. [Forward].
11. Select [Start] button.

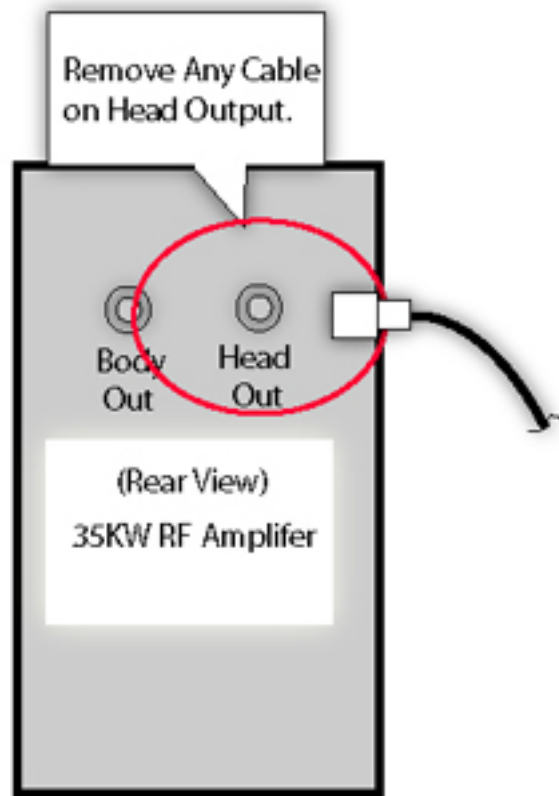
NOTE: Advisory message indicating head coil not present can be ignored.

12. Status should indicate Pass/Fail from 1 minute to 7.5 minutes depending on software version (tool should notify user of current step in progress). If Failure occurs, see [Universal Power Monitor \(UPM\) Troubleshooting](#).
13. Proceed to [UPM- Body and Head Functional Check](#) before moving to [Finalization, Step 1](#).

4.3 HEAD REFLECTED POWER CALIBRATION

1. Disconnect the RF Dummy Load from the Head Output J3. (see [Illustration 5](#)).

Illustration 5: Remove Any Cables From Head RF Out



2. Disconnect the RF Dummy Load from the Head Output J3 on the 25 KW amp. See [Illustration 6](#).

NOTE: If the amp at your site is a 25kW/35kW version with an LCD on the front, disconnect the Dummy Load from J3 directly as there are no RF couplers required for that amplifier. See [Illustration 7](#).

Illustration 6: Remove Any Cables From Head RF Out 25KW Amp

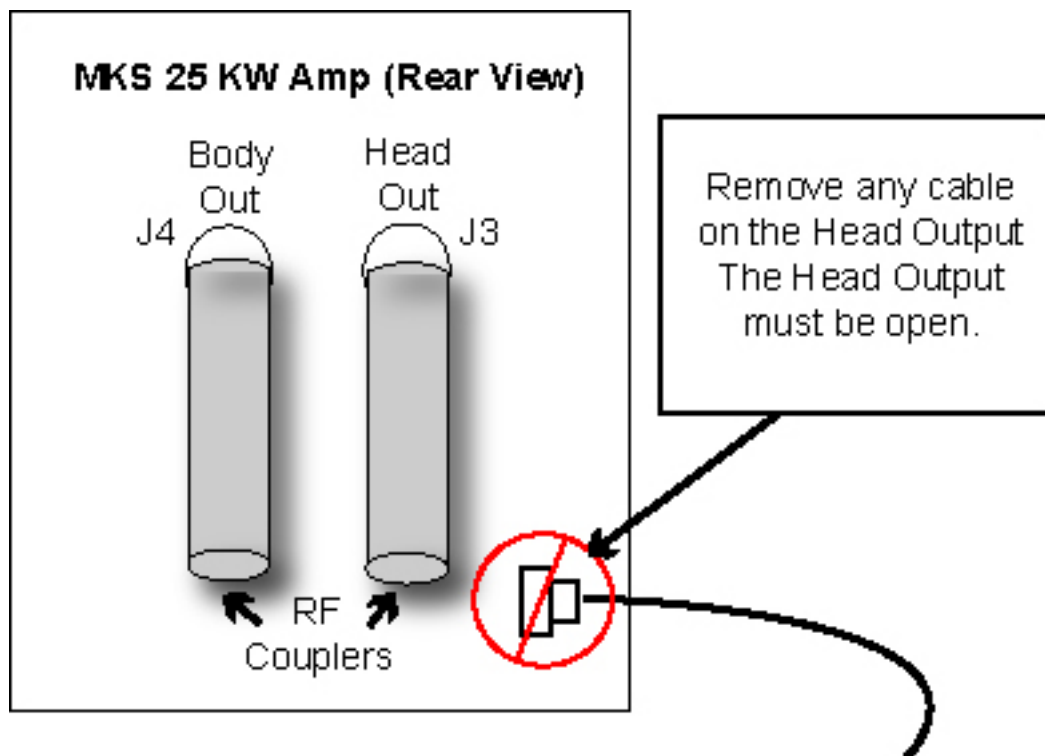
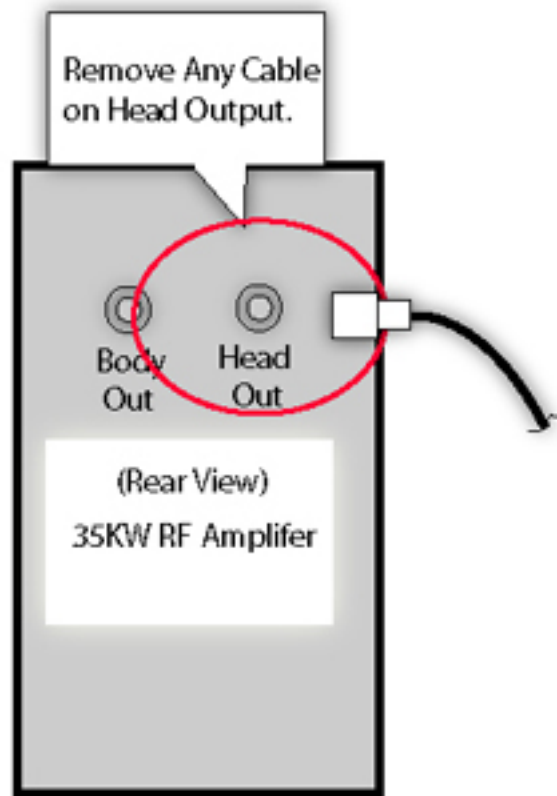


Illustration 7: Remove Any Cables From Head RF Out 25kW/35kW Amp



3. Locate the cable stub in the power measurement kit and connect it to j3 on the 35 KW amplifier and the 25/35 KW amplifier. DO NOT connect it to the 25 KW amplifier.
4. Return to the UPM Tool User Interface.
5. Make sure Nuclei [1H] is selected.
6. Make sure [Head] is selected.
7. Select Power Monitor type. [Reflected].
8. Make sure [Calibrate] is selected.
9. Select the [Start] button.
10. In 1 minute to 7.5 minutes depending on software version (tool should notify user of current step in progress and displays Pass/Fail. If Failure occurs, see [Universal Power Monitor \(UPM\) Troubleshooting](#).

5 Finalization

1. Remove the cables and power measurement equipment.
2. Re-attach the Head and Body Heliax Cables.
3. Remove the cables and power measurement equipment.
4. Re-attach the Head Heliax Cables.
5. Re-enable T/R Drive fault detection. See [Illustration 1](#).
6. Upon completion of any UPM calibration, [Reset TPS]. The system will not scan until this step is complete.